

Nested Queries & Subqueries

Q. What are Nested Queries (Subqueries) in SQL Differentiate between Single-Row, Multiple-Row, and Correlated subqueries with examples.

[DEF] Definition to Remember

A Nested Query (or Subquery) is a complete SQL query placed inside another SQL query (the outer query). The inner query executes first and provides data to the outer query. Subqueries can be placed in WHERE, HAVING, or FROM clauses.

1. Types of Subqueries

A. Single-Row Subquery

The inner query returns one value (e.g., 45000). The outer query uses it.

B. Multiple-Row Subquery

C. Correlated Subquery

2. Comparison of Subquery Types

Feature	Single-Row	Multiple-Row	Correlated
Rows Returned	Exactly 1	More than 1	Varies per outer row
Operators	`=`, `>`, `<`	`IN`, `ANY`, `ALL`	`=`, `>`, `<`
Execution	Once	Once	Once per outer row
Performance	Fast	Moderate	Slow (re-executes per row)

[MUST-WRITE] Must-Write Points (for 10 marks)

1. A Subquery/Nested Query is a SELECT query placed inside another SQL query.
2. The inner query executes first and provides data to the outer query.
3. Single-Row Subquery: returns one value; uses operators like =, >, <.
4. Multiple-Row Subquery: returns multiple rows; must use IN, ANY, or ALL.

5. Correlated Subquery: inner query references outer query -- re-runs for every outer row.

6. Correlated subqueries are more powerful but slower (expensive execution).

7. Subqueries can be placed in WHERE, HAVING, or FROM clauses.

[RECALL] Quick Recall

Subquery (inner executes first) -> Single-Row (=,>,< operators) -> Multi-Row (IN/ANY/ALL) -> Correlated (references outer, re-runs per row, expensive)